



College Management System

Transforming Traditional Campuses into AI-Enabled Smart Campuses.

"Empowering minds, automating administrative workflows, and building the future of campus governance."

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Here
AI * Automation * Analytics *
Governance**

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MODULE 01

ADMISSION MANAGEMENT

SYSTEM TYPE
100% Online Portal

SEAT ALLOCATION
Cutoff Engine

STUDENT ID
Auto-Generated

1. Key Sub-Modules & Core Capabilities

Online Application Portal | Quota Matrix | Document Verification | Fee Receipt Sync | Seat Allotment Engine | Roll Number Allocation | Branch Placement

2. Operational Features & Automation Rules

- [+] Interactive student application portal with real-time validation checks for document uploads.
- [+] Merit list calculation and automated quota screening (General / Management / NRI).
- [+] Automatic student profile creation and institutional registration number generation upon fee payment.

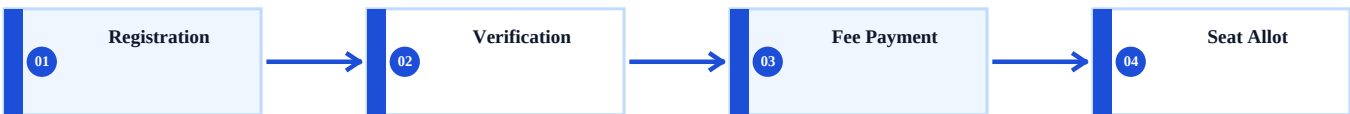
3. Working Process & Workflow Execution

Prospects register online and upload certificates. System validates cutoff percentages, assigns document verification slots, records quota reservations, and locks seats upon fee payment receipt. Finally, branch allocations and roll numbers generate.

4. Role-Based Access Controls (RBAC)

Authorized System Role	Operational Scope & Governance Permissions
Super Admin / Admins	Full administrative control. Modify cutoff rules, override quota reservations, and approve document waivers.
Principal / Dean	Approve final seat allocation lists, view merit rank logs, and monitor intake analytics.
Student Applicant	Fill admission forms, upload certificates, pay registration fees, and track application status.

5. System Operational Workflow Flowchart



MODULE 02

STUDENT INFORMATION SYSTEM (SIS)

PROFILE RECORD
360° Digital Card

CERTIFICATES
Instant Generator

ALUMNI BRIDGE
Auto-Graduation

1. Key Sub-Modules & Core Capabilities

Student 360 Card | Academic History | Certificate Generator | Parent Contact Sync | ID Card Printer | Discipline Register | Graduation Audit

2. Operational Features & Automation Rules

- [+] Central 360-degree digital master profile card for every enrolled student.
- [+] Automated certificate generator (Bonafide, Transfer, Conduct, Course Completion).
- [+] Continuous record tracking that seamlessly archives profiles into the Alumni database upon graduation.

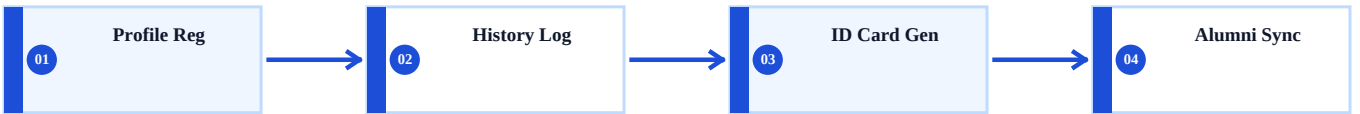
3. Working Process & Workflow Execution

Students input profile details and upload photo proofs. The system formats printable digital ID cards, maps academic semester records, tracks disciplinary logs, and updates parent details. On graduation, profiles auto-transfer to the Alumni database.

4. Role-Based Access Controls (RBAC)

Authorized System Role	Operational Scope & Governance Permissions
Super Admin / Admins	Full read/write rights. Configure certificate templates and manage disciplinary records.
HOD / Faculty	View student profile cards, monitor academic progression, and access parent contact details.
Student / Parents	Read-only access to personal profiles, grade cards, and address change submission.

5. System Operational Workflow Flowchart



MODULE 03

ACADEMIC MANAGEMENT

CURRICULUM
CBCS Regulation

ELECTIVES
Merit Auto-Alloc

TIMETABLE
Clash-Free Engine

1. Key Sub-Modules & Core Capabilities

Academic Calendar | Semester Structuring | Course Registration | Curriculum Regulations | Credit Banking | Open Electives | Visual Timetables

2. Operational Features & Automation Rules

- [+] Configurable curriculum regulations (Choice Based Credit System - CBCS / R23 / R20).
- [+] Automatic elective course allocation engine based on student preference and merit ranks.
- [+] Clash-free visual timetabling matrix mapping classroom capacity and faculty availability.

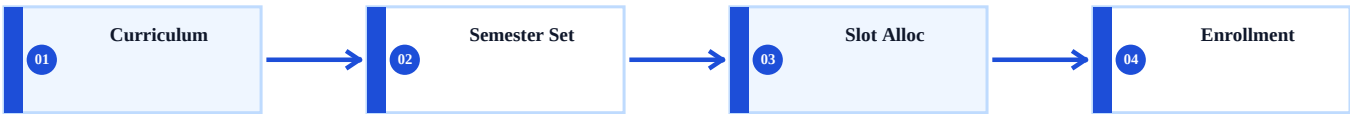
3. Working Process & Workflow Execution

Deans create semesters and define course codes under regulations. Faculty submit weekly lecture availability. The system compiles visual timetables. Students log in during course registration windows to select open electives.

4. Role-Based Access Controls (RBAC)

Authorized System Role	Operational Scope & Governance Permissions
Super Admin / Deans	Create semesters, regulations, configure course credits, and publish academic calendars.
HOD / Faculty	Upload syllabus blueprints, verify course mappings, and manage lab allocations.
Student	Select open electives and complete semester course registrations.

5. System Operational Workflow Flowchart



MODULE 04

ATTENDANCE

TRACKING MODE
Biometric / RFID

SHORTAGE ALERT
75% Threshold

PARENT ALERT
Automated SMS

1. Key Sub-Modules & Core Capabilities

Biometric RFID Sync | Daily Class Register | 75% Shortage Tracker | Parent SMS Trigger | Duty Leave Approvals | Department Analytics

2. Operational Features & Automation Rules

- [+] Real-time biometric RFID hardware integration pulling turnstile punch logs directly to DB.
- [+] Automated attendance shortage calculation flags students below the mandatory 75% threshold.
- [+] SMS messaging engine dispatching immediate absentee alerts to registered parent mobile numbers.

3. Working Process & Workflow Execution

Biometric devices capture entry logs. Faculty review and verify class sheets. The shortage calculator evaluates weekly compliance scores, generates warning letters, and triggers SMS alerts to parents.

4. Role-Based Access Controls (RBAC)

Authorized System Role	Operational Scope & Governance Permissions
Super Admin / Admin	Configure biometric ports, clear hardware buffers, and approve attendance waivers.
HOD / Faculty	Submit daily class attendance sheets, review department percentages, and sign duty leaves.
Student / Parent	View monthly attendance percentages, calendar check-in logs, and warning notices.

5. System Operational Workflow Flowchart



MODULE 05

EXAMINATION (CONTROLLER SUITE)

EVALUATION
SGPA/CGPA Engine

SECURITY
AES-256 Lock

REVALUATION
Instant Grade Sync

1. Key Sub-Modules & Core Capabilities

Exam Schedules | Hall Ticket Clearance | Encrypted Question Bank | Exam Centers | Invigilator Roster | Seating Matrix | Grace Marks Board | Malpractice Verdicts

2. Operational Features & Automation Rules

- [+] Dedicated Controller of Examinations (COE) Governance Suite with outcome-based SGPA/CGPA computation.
- [+] Automated Hall Ticket clearance and withhold controls for fee defaulters or low attendance (<65%).
- [+] AES-256 Question Paper encryption, invigilation duty rosters, and anti-malpractice seating matrix.
- [+] Board moderation, revaluation grade updates, grace marks (+1/+2) sanctioning, and malpractice verdicts.

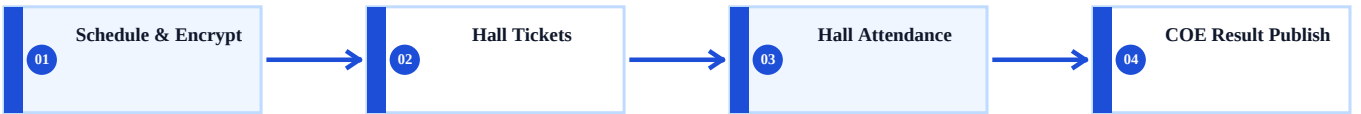
3. Working Process & Workflow Execution

Exam branch creates timetables and allocates centers. Candidates clear fee/attendance checks to unlock hall tickets. Question papers undergo AES-256 encryption. Hall invigilators track live attendance. Internal and external marks are moderated, results process to SGPA/CGPA, and grade cards publish.

4. Role-Based Access Controls (RBAC)

Authorized System Role	Operational Scope & Governance Permissions
Controller of Examinations (COE) / Dean	Complete administrative authority: Approve results, sanction grace marks, execute malpractice verdicts, lock encrypted question papers.
HOD / Faculty	Upload internal assessment marks, confirm invigilation duties, and verify moderation keys.
Student	Register for exams, download hall tickets, view SGPA/CGPA results, and apply for revaluation.

5. System Operational Workflow Flowchart



MODULE 06

FACULTY ERP

WORKLOAD
Weekly Tracker

RESEARCH LOG
Scopus & NAAC

APPRAISAL
Self-Assessment

1. Key Sub-Modules & Core Capabilities

Faculty Profile | Workload Manager | Leave Portal | Attendance Register | Research Publications | Grant Tracker | Performance Appraisal | Payroll Ledger

2. Operational Features & Automation Rules

- [+] Workload manager tracking weekly lecture, tutorial, and lab hours against statutory guidelines.
- [+] Research and publication repository cataloging Scopus, Web of Science, and UGC journal papers.
- [+] Self-appraisal digital sheets linked directly to NAAC/NBA accreditation criteria.

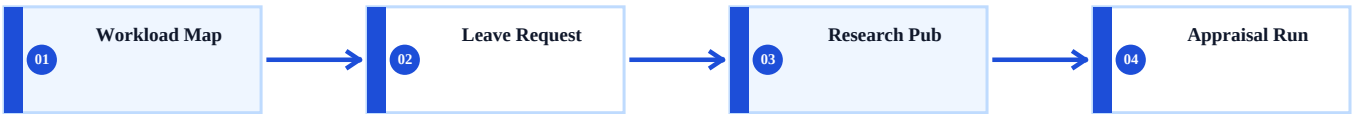
3. Working Process & Workflow Execution

Faculty update profiles, upload research citations, submit leave requests, and track weekly lecture schedules. Department heads review performance ratings, calculate appraisal scores, and approve leave allocations.

4. Role-Based Access Controls (RBAC)

Authorized System Role	Operational Scope & Governance Permissions
Super Admin / Admin	Approve promotions, modify salary scale bands, and configure teaching workload limits.
HOD / Dean	Review faculty workloads, approve research grant requests, and sanction leaves.
Faculty	Manage personal profiles, submit leave applications, upload research publications, and view payslips.

5. System Operational Workflow Flowchart



MODULE 07

LEARNING MANAGEMENT SYSTEM

CONTENT HUB
PDF / MP4 / Docs

ASSESSMENT
Online Quiz Key

SUBMISSION
Plagiarism Check

1. Key Sub-Modules & Core Capabilities

Study Materials | Video Lectures | Digital Assignments | Online Quiz Engine | Discussion Forum | Virtual Classroom | Gradebook Sync

2. Operational Features & Automation Rules

- [+] Central study material repository supporting PDF notes, video lectures, and presentation decks.
- [+] Automated online quiz modules featuring randomized question banks and instant key grading.
- [+] Assignment submission portal with deadline enforcement and plagiarism check indicators.

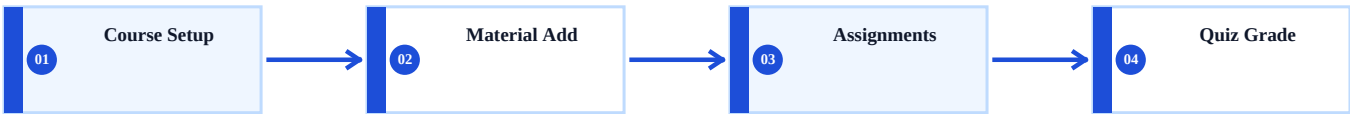
3. Working Process & Workflow Execution

Faculty create course modules, upload study files, and schedule timed quizzes. Students access study materials, watch video lectures, submit assignments, and participate in discussion threads. Quiz scores auto-sync to gradebooks.

4. Role-Based Access Controls (RBAC)

Authorized System Role	Operational Scope & Governance Permissions
HOD / Faculty	Upload course materials, create online quizzes, evaluate assignments, and monitor forums.
Student	Download study notes, stream video lectures, submit assignments, and attempt online quizzes.

5. System Operational Workflow Flowchart



MODULE 08

PLACEMENT CELL

ELIGIBILITY
CGPA / Backlog Filter

RECRUITER PORTAL
Job & Drive Post

ANALYTICS
Package & Highest

1. Key Sub-Modules & Core Capabilities

Company Registry | Job Drive Postings | Eligibility Screener | Student Applications | Aptitude Testing | Interview Schedule | Offer Letter Vault | Placement Analytics

2. Operational Features & Automation Rules

- [+] Automated eligibility criteria filter screening candidates by minimum CGPA and active backlog limits.
- [+] Corporate recruiter portal for posting vacancies, filtering candidate resumes, and managing drive schedules.
- [+] Placement analytics dashboard visualizing branch-wise placement percentages, average packages, and top recruiters.

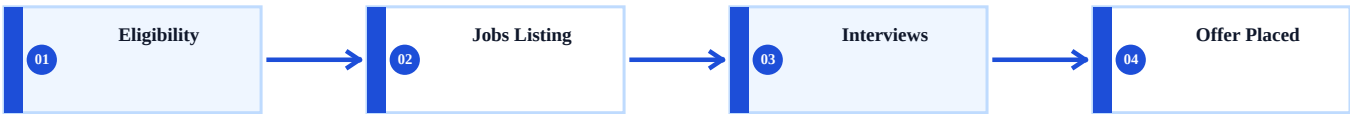
3. Working Process & Workflow Execution

Placement officer registers corporate recruiters. Companies post vacancies with criteria. Eligible students apply. System schedules online aptitude tests and interview panels. Selected candidates receive digital offer letters.

4. Role-Based Access Controls (RBAC)

Authorized System Role	Operational Scope & Governance Permissions
Placement Officer (TPO)	Register corporate profiles, set CGPA eligibility thresholds, schedule drives, and log offer letters.
Recruiter	Post job drives, review filtered student profiles, and update candidate hiring status.
Student	View active placement drives, check eligibility status, submit applications, and download offer letters.

5. System Operational Workflow Flowchart



MODULE 09

HOSTEL MANAGEMENT

ROOM MATRIX
Live Vacancy Map

ATTENDANCE
Night Check-in

PASS SYSTEM
Digital Outing

1. Key Sub-Modules & Core Capabilities

Room Allocation | Night Attendance | Hostel Fee Ledger | Visitor Registry | Outing Gatepasses | Maintenance Complaints | Mess Menu Manager

2. Operational Features & Automation Rules

- [+] Live bed matrix displaying vacant, occupied, and maintenance-blocked room allocations in real time.
- [+] Digital night attendance register with mobile biometric check-in integration for wardens.
- [+] Automated complaint ticketing system routing plumbing, electrical, and mess issues to maintenance staff.

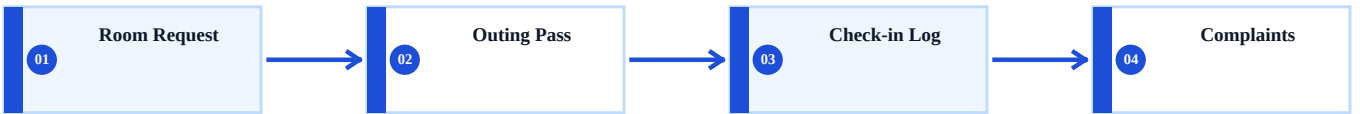
3. Working Process & Workflow Execution

- Students apply for hostel rooms online. System verifies fee clearance and allocates bed numbers. Wardens record daily night attendance.
- Outing gatepasses require warden approval and security gate validation.

4. Role-Based Access Controls (RBAC)

Authorized System Role	Operational Scope & Governance Permissions
Hostel Warden	Allocate rooms, verify night attendance logs, approve outing passes, and resolve maintenance tickets.
Accounts Manager	Post hostel fees, track payment dues, and issue clearance receipts.
Student Resident	Apply for room allocation, submit outing requests, log maintenance complaints, and rate mess menus.

5. System Operational Workflow Flowchart



MODULE 10

TRANSPORT MANAGEMENT

TRACKING
Live GPS Coordinates

ROUTES
Stop & Capacity Map

PASS CONTROL
QR / RFID Pass

1. Key Sub-Modules & Core Capabilities

Route Management | Bus Capacity Map | Live GPS Tracking | Driver Allocations | Transport Fee Ledger | Digital Bus Passes | Vehicle Maintenance

2. Operational Features & Automation Rules

- [+] Route mapper tracking bus stops, fare zones, and seating capacities across fleet vehicles.
- [+] Live GPS vehicle location integration feeding real-time coordinates to student/parent dashboards.
- [+] Digital transport fee collection register linked directly to automated bus boarding pass issuance.

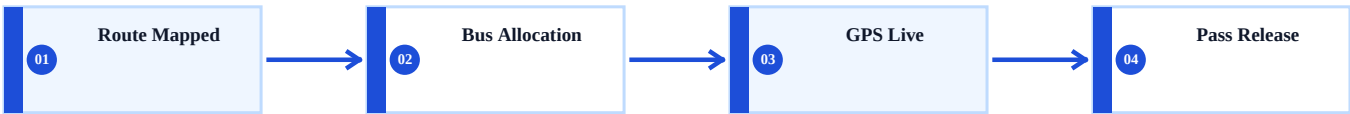
3. Working Process & Workflow Execution

Transport manager defines bus routes, pick-up stops, and assigns drivers. Students select routes and pay transport fees online. Digital bus passes generate, and live GPS coordinates stream during transit hours.

4. Role-Based Access Controls (RBAC)

Authorized System Role	Operational Scope & Governance Permissions
Transport Officer	Define routes, assign vehicles and drivers, monitor maintenance schedules, and audit fleet logs.
Accounts Cell	Manage transport fee receipts and issue digital bus boarding passes.
Student / Parent	View bus route schedules, track live GPS locations on mobile devices, and download bus passes.

5. System Operational Workflow Flowchart



MODULE 11

LIBRARY MANAGEMENT

SEARCH ENGINE
OPAC Catalog

TRACKING
Barcode / RFID

FINE CALCULATOR
Auto-Dues Sync

1. Key Sub-Modules & Core Capabilities

OPAC Book Catalog | Barcode Issue / Return | Overdue Fine Calculator | Digital E-Books | Journal Subscription | Member Registry | Book Reservation

2. Operational Features & Automation Rules

- [+] Online Public Access Catalog (OPAC) index for physical books, journals, and digital research papers.
- [+] Barcode/RFID scanner integration executing check-outs and returns within seconds.
- [+] Automated library fine calculation engine linked directly to student financial fee ledgers.

3. Working Process & Workflow Execution

Librarians scan book barcodes for check-outs. System monitors return due dates and sends overdue reminders. Overdue fines auto-calculate daily and sync to the student fee ledger.

4. Role-Based Access Controls (RBAC)

Authorized System Role	Operational Scope & Governance Permissions
Librarian	Manage book catalog, scan barcodes, override library fines, upload e-books, and monitor circulation.
Student / Faculty	Search OPAC catalog, reserve books online, view checkout history, access e-books, and pay fines.

5. System Operational Workflow Flowchart



MODULE 12

FINANCE & ACCOUNTING

LEDGER Tuition & Fee Dues	SCHOLARSHIPS Scheme Adjustments	AUDITING GST & Balance Sheet
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1. Key Sub-Modules & Core Capabilities

Tuition Fee Ledger Online Payment Gateway Scholarship Adjustments Staff Payroll Disbursements Vendor Invoices GST Compliance Financial Balance Sheets

2. Operational Features & Automation Rules

- [+] Tuition fee ledger managing online payment gateways, installment plans, and transaction receipts.
- [+] Scholarship adjustment engine mapping government schemes, merit concessions, and fee waivers.
- [+] Automated balance sheet, P&L statement, TDS filing, and GST compliance report generator.

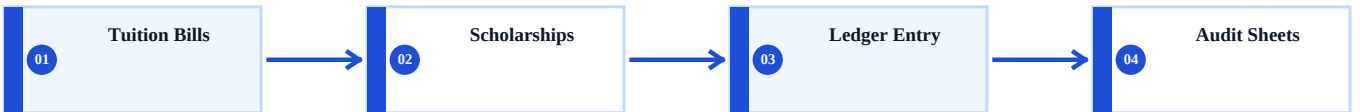
3. Working Process & Workflow Execution

Finance department generates fee structures. Students pay fees online via payment gateways. Accounts verify scholarship grants, process vendor bills, execute staff payroll disbursements, and generate annual audit ledgers.

4. Role-Based Access Controls (RBAC)

Authorized System Role	Operational Scope & Governance Permissions
Super Admin / Finance Head	Full financial control: Modify fee structures, approve vendor payments, generate balance sheets.
Accounts Team	Collect tuition fees, verify scholarship adjustments, disburse staff salaries, and file GST returns.
Student / Parent	View fee balance ledgers, make online tuition payments, and download official fee receipts.

5. System Operational Workflow Flowchart



MODULE 13

HRMS & STAFF PAYROLL

PERSONNEL
Digital Staff Files

ATTENDANCE
Biometric Sync

PAYROLL
Scale & Allowances

1. Key Sub-Modules & Core Capabilities

Employee Directory | Recruitment Portal | Biometric Attendance Sync | Leave Approval Workflow | Salary Scale Engine | Pay Slip Generator | Service History Log

2. Operational Features & Automation Rules

- [+] Central employee digital repository storing qualifications, service history, and statutory documents.
- [+] Biometric RFID check-in sync evaluating working hours, overtime, and leave deductions automatically.
- [+] Automated payroll disbursement engine calculating basic pay, DA, HRA, PF, TDS, and generating PDF payslips.

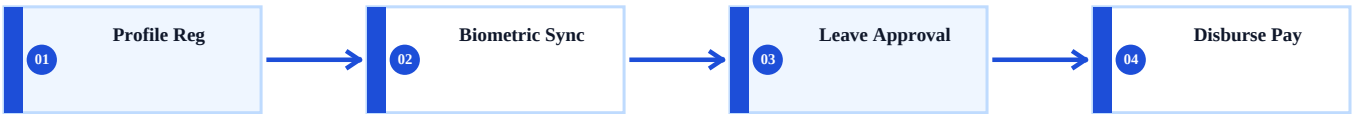
3. Working Process & Workflow Execution

HR creates employee profiles. Biometric hardware syncs daily staff attendance. Department heads approve leaves. Payroll engine computes monthly salaries, applies tax/PF deductions, and generates digital payslips.

4. Role-Based Access Controls (RBAC)

Authorized System Role	Operational Scope & Governance Permissions
Super Admin / HR Manager	Create employee records, configure salary bands, approve leaves, and disburse payroll.
Accounts Officer	Manage tax deductions, process PF ledgers, and execute bank salary transfer files.
Faculty / Staff	View personal service profile, submit leave requests, track biometric logs, and download payslips.

5. System Operational Workflow Flowchart



MODULE 14

INVENTORY & ASSET CONTROL

ASSET LEDGER
WDV Depreciation

STOCK ALERT
Low Volume Warning

SECURITY
Gatepass Audit

1. Key Sub-Modules & Core Capabilities

Asset Master Register | Lab Equipment Log | Purchase Order (PO) | Consumables Stock | Depreciation Calculator | Maintenance Tickets | Outward Gatepasses

2. Operational Features & Automation Rules

- [+] Asset register tracking initial capital costs, location assignments, and Written Down Value (WDV) depreciation.
- [+] Stock consumable indicator raising automatic warnings when lab supplies drop below safety thresholds.
- [+] Outward gatepass log tracking equipment sent out for servicing, calibration, or inter-campus transfer.

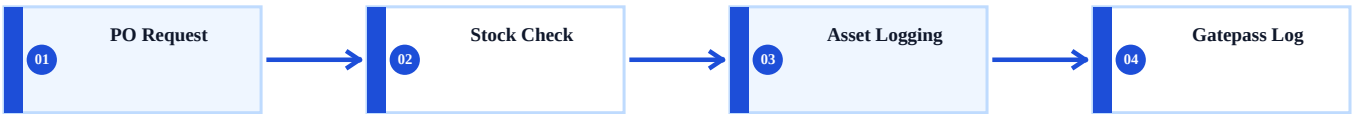
3. Working Process & Workflow Execution

Department custodians log equipment details. Purchase requests are approved. Stock levels auto-adjust on item issuance. Maintenance tickets trigger repair schedules, and gatepasses authorize security checks.

4. Role-Based Access Controls (RBAC)

Authorized System Role	Operational Scope & Governance Permissions
Super Admin / Admin	Full asset ledger authority: Approve purchase orders, authorize gatepasses, and write off assets.
Accounts	Verify purchase order invoices, update asset ledger values, and execute stock inventory audits.
HOD / Lab Custodian	Log equipment status, request consumables restocking, and create maintenance tickets.

5. System Operational Workflow Flowchart



MODULE 15

ACCREDITATION (NBA / NAAC)

NAAC SSR
Criteria 1-7 Vault

NBA OBE
CO-PO Attainment

CERTIFICATES
Auto Generator

1. Key Sub-Modules & Core Capabilities

NAAC Criteria Repository | NBA CO-PO Attainment | NIRF / AISHE Exporter | Event Proposals | Participant Check-in | Evidence Vault | Digital Certificate Signer

2. Operational Features & Automation Rules

- [+] NAAC Self-Study Report (SSR) evidence repository supporting criteria-wise document categorization.
- [+] NBA Outcome-Based Education (OBE) matrix mapping Course Outcomes (CO) to Program Outcomes (PO).
- [+] Event workflow engine managing approval flows, attendance verification, and auto-signed digital certificates.

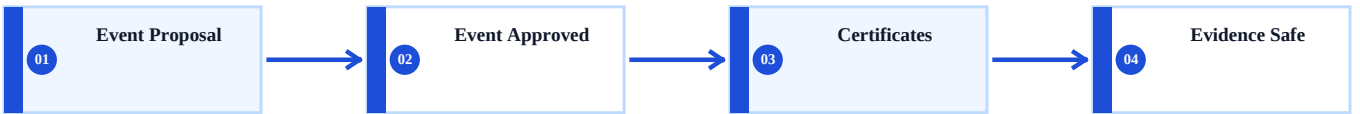
3. Working Process & Workflow Execution

Faculty submit event proposals. HOD and Principal approve. Students register and attend. Event coordinators upload reports and photo proofs into the accreditation vault. System generates signed participant certificates.

4. Role-Based Access Controls (RBAC)

Authorized System Role	Operational Scope & Governance Permissions
Super Admin	Manage accreditation parameters, export NAAC/NBA datasets, and oversee evidence repositories.
Principal / Dean	Approve event proposals, verify accreditation criteria files, and authorize final SSR submissions.
HOD / Coordinator	Map CO-PO metrics, upload department evidence files, mark event attendance, and issue certificates.
Student	Register for academic events, check-in to sessions, and download verified participation certificates.

5. System Operational Workflow Flowchart



MODULE 16

COMMUNICATION & CIRCULARS

CHANNELS

SMS / Email / Push

BROADCASTING

Instant Dispatch

ARCHIVE

Public Circulars

1. Key Sub-Modules & Core Capabilities

Multi-Channel Broadcast | Official Circulars | Parent SMS Engine | Email Newsletter | Push Notifications | Notice Board Portal | Delivery Logs

2. Operational Features & Automation Rules

- [+] Multi-channel broadcast engine dispatching announcements via SMS, Email, Push Notifications, and WhatsApp.
- [+] Official circular archive displaying verified administrative orders on role dashboards.
- [+] Automated trigger notifications for fee payment reminders, exam dates, emergency holidays, and event updates.

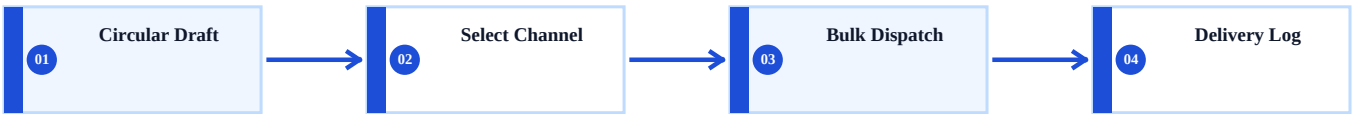
3. Working Process & Workflow Execution

Authorized staff draft announcements and select target audiences (Students / Parents / Staff). Dispatch engine transmits bulk messages across channels while simultaneously publishing circulars to user dashboards.

4. Role-Based Access Controls (RBAC)

Authorized System Role	Operational Scope & Governance Permissions
Super Admin / Admins	Send institution-wide notifications, publish official circulars, and monitor delivery logs.
HOD / Faculty	Message department students, broadcast class announcements, and issue academic notices.
Student / Parents	Read-only access to circular notice boards and notification inbox feeds.

5. System Operational Workflow Flowchart



MODULE 17

GRIEVANCE & REDRESSAL

PRIVACY

100% Anonymous Mode

COMPLIANCE

Anti-Ragging Vault

TRACKING

Resolution Tickets

1. Key Sub-Modules & Core Capabilities

Anonymous Complaint Box | Anti-Ragging Committee | Women Cell Portal | Case Assignment | Investigation Log | Hearing Schedule | Resolution Audit

2. Operational Features & Automation Rules

- [+] 100% anonymous complaint submission option preserving student and staff privacy during sensitive filings.
- [+] Legal compliance trackers for Anti-Ragging, Internal Complaints Committee (ICC), and Women Cell regulations.
- [+] Resolution ticketing system tracking case assignments, hearing dates, investigation notes, and final orders.

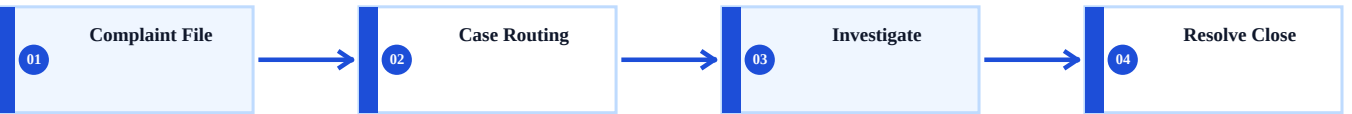
3. Working Process & Workflow Execution

Complainant registers a grievance online. System assigns a confidential tracking ticket. Designated committee reviews details, conducts hearings, logs investigation notes, and issues official resolution orders.

4. Role-Based Access Controls (RBAC)

Authorized System Role	Operational Scope & Governance Permissions
Grievance Committee	Review filed complaints, assign investigators, record hearing notes, and issue resolution orders.
Super Admin	Monitor grievance resolution statistics, verify legal compliance, and audit pending case logs.
Student / Staff	Submit grievances (anonymous or named), upload supporting evidence, and track ticket status.

5. System Operational Workflow Flowchart



MODULE 18

ALUMNI RELATIONS

DIRECTORY
Verified Alumni Log

MENTORSHIP
Student Match

DONATIONS
80G Tax Receipt

1. Key Sub-Modules & Core Capabilities

Alumni Directory | Career Job Portal | Mentorship Network | Reunion Events | Donation Gateway | Success Stories | 80G Tax Receipts

2. Operational Features & Automation Rules

- [+] Verified alumni directory automatically populated from student graduation database records.
- [+] Mentorship network pairing experienced alumni professionals with current students for career guidance.
- [+] Secure donation gateway issuing official tax benefit receipts (80G) for institutional contributions.

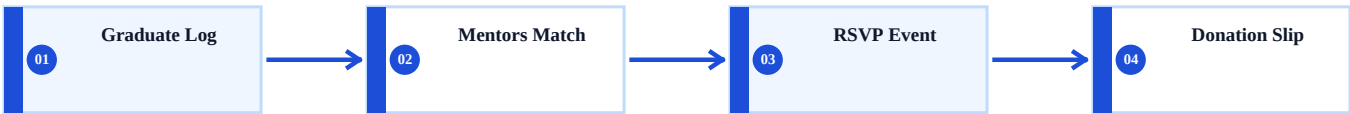
3. Working Process & Workflow Execution

Graduating students transition to the alumni directory. Alumni update employment details, post job referral openings, offer mentorship slots, and make institutional donations. System issues automated tax receipts.

4. Role-Based Access Controls (RBAC)

Authorized System Role	Operational Scope & Governance Permissions
Alumni Officer / Admin	Verify alumni directory records, organize reunion events, and manage donation ledgers.
Alumni	Update professional profiles, post job/internship openings, offer mentorship, and donate funds.
Student	Browse alumni directory, apply for alumni job referrals, and request mentorship pairings.

5. System Operational Workflow Flowchart



MODULE 19

ADMINISTRATION & SECURITY

RBAC

19-Module Matrix

AUDIT LOG

Immutable Tracking

DB BACKUP

Automated Schedules

1. Key Sub-Modules & Core Capabilities

Role Permissions Matrix | Unified User Control | Immutable Audit Logs | System Configuration | Security Hardening | Automated DB Backups | Global Analytics

2. Operational Features & Automation Rules

- [+] Granular Role-Based Access Control (RBAC) engine configuring permissions across all 19 system modules.
- [+] Immutable audit trail logging every user login, database modification, API request, and export action.
- [+] Automated database backup cron-jobs creating daily encrypted snapshots with single-click restore capabilities.

3. Working Process & Workflow Execution

Super admins configure role permissions and assign user credentials. The system logs all user actions in an immutable audit ledger. Nightly cron-jobs create encrypted database backups and monitor system health.

4. Role-Based Access Controls (RBAC)

Authorized System Role	Operational Scope & Governance Permissions
Super Admin	Complete system control: Configure RBAC matrix, manage system settings, view audit logs, schedule backups.
Auditors / Deans	View system health metrics, monitor user activity audit logs, and export governance reports.

5. System Operational Workflow Flowchart

